NATIONAL AVIATION ACADEMY OF THE PHILIPPINES • INSTITUTE OF COMPUTER STUDIES

Manuscript vs. Presentation Alignment Audit & Defense Guide

A Web-Based Inventory and Point-of-Sale (POS) System with Data Analytics for Zoe Pharmacy and General Merchandise

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DEGREE & CAMPUS
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DOCUMENTS AUDITED
146-Page PDF Manuscript vs. 18 Google Slides

DEFENSE DATE
October 2026

100%
Architecture & Methodology
Waterfall, React/PHP/PostgreSQL & FIFO/FEFO fully synced.

68.95%
Financial ROI (Identical)
CapEx ₱85,800, NPV ₱50,878, 1.78-yr payback exact match.

0.83%
Inventory Discrepancy
Exact reduction from 17.32% across both documents.

3 Items
Slide Discrepancies
Older draft figures on Slides 8, 9, 11, 13, and 16 require sync.

1. Master Slide-by-Slide to Manuscript Mapping

Every slide in your 18-slide deck was cross-referenced with the exact chapters, pages, tables, and figures of your 146-page manuscript:

SLIDE #	SLIDE TITLE	MANUSCRIPT CHAPTER & PAGE	ALIGNMENT STATUS	VERIFICATION SUMMARY
Slide 1	Title Slide	Page 1 (Title Page)	**100% Match**	Title, authors, NAAP ICS, BS Aviation IT, and October 2026 date align perfectly.
Slide 2	Project Context: Manual to Unified System	Ch. 1 (pp. 2–4, 6–8); Ch. 2 (pp. 32–33)	**100% Match**	Dual-operator setup (Margarita & Inigo), logbook delays, unmonitored spoilage, and retail-pharmacy compliance gap.
Slide 3	Research Focus & Objectives	Ch. 1 (pp. 4–5 [RQs]; pp. 11–12 [Objectives])	**100% Match**	Word-for-word alignment with the five (5) research questions and specific objectives.
Slide 4	Proposed System: Core Features	Ch. 1 (pp. 8–9); Ch. 3 (pp. 41–42, 67–68, 75–77)	**100% Match**	Covers Inventory, POS, FIFO/FEFO rules, color-coded alerts, analytics dashboard, and printable PDF audit reports.
Slide 5	System Architecture & Workflow	Ch. 3 (pp. 34–35, 70–73, 78–84, Figs 3–6)	**100% Match**	3-Tier Thick Client (React.js/Vite frontend, PHP RESTful API, PostgreSQL DB), Context Diagram, and DFD Level 0.
Slide 6	Demand Forecasting: SMA vs. Exponential Smoothing	Ch. 1 (p. 3); Ch. 2 (pp. 28–30); Ch. 3 (pp. 73–75)	**100% Match**	Rationale for $\alpha = 0.7$ smoothing constant over 3-day static SMA to absorb volatile local demand shocks.
Slide 7	Research Methodology	Ch. 4 (pp. 94–102)	**100% Match**	Quasi-experimental one-group design; 10 IT experts (ISO 25010), 2 store owners (TAM), 50 simulated sales, 23-day data window.

SLIDE #	SLIDE TITLE	MANUSCRIPT CHAPTER & PAGE	ALIGNMENT STATUS	VERIFICATION SUMMARY
Slide 8	Reported Results & Key Findings	Ch. 4 (pp. 118–123); Ch. 5 (pp. 126–129)	Discrepancy	Slide uses older values for checkout speed (84.15s vs 37.38s) and forecasting error metrics (2.92 vs 0.54).
Slide 9	Objective 1 — Operational Impact	Ch. 4 (p. 119, Table 30); Ch. 5 (p. 126, 129)	Discrepancy	Slide lists digital checkout at 84.15s (60.07% reduction). Manuscript Table 30 records **37.38s (82.26% reduction)**.
Slide 10	Objective 2 — Inventory Control	Ch. 4 (p. 120, Table 31); Ch. 5 (p. 126, 130)	100% Match	Baseline physical discrepancy of 17.32% (15/120 SKUs) reduced to 0.83% post-implementation. Identical.
Slide 11	Objective 3 — Forecasting Accuracy	Ch. 4 (pp. 121–122, Table 32); Ch. 5 (pp. 127, 130–131)	Critical Discrepancy	Slide claims ES won (MAPE 8.40% vs 22.85%). Manuscript Table 32 proves SMA had lower error (MAE 0.52 vs 0.54); H1 was NOT supported.
Slide 12	Objective 4 — User Acceptance (TAM)	Ch. 4 (pp. 113–118, Tables 25–29); Ch. 5 (p. 128)	100% Match	Composite Mean = 3.68/4.00 (PU = 3.90, PEOU = 3.80, Attitude = 3.70, Intention = 3.50, Actual Use = 3.50).
Slide 13	Objective 5 — Software Quality (ISO 25010)	Ch. 4 (pp. 104–113, Tables 16–24); Ch. 5 (p. 128)	Minor Discrepancy	Grand Mean = 3.60 across both. Slide lists Security at 3.26; Manuscript Table 21 lists Security at **3.14** (Admin override = 2.30).
Slide 14	Scope, Limitations & Practical Value	Ch. 1 (pp. 14–18)	100% Match	Scope boundaries: single micro-enterprise, no BIR official receipts, no GCash API, no native app, 23-calendar-day dataset.

SLIDE #	SLIDE TITLE	MANUSCRIPT CHAPTER & PAGE	ALIGNMENT STATUS	VERIFICATION SUMMARY
Slide 15	Financial Feasibility (CBA Model)	Ch. 3 (pp. 60–65, Tables 10–13); Ch. 5 (p. 129)	100% Match	CapEx ₱85,800; Tangible Annual ₱48,320; 3-Yr Benefit ₱144,960; ROI 68.95%; NPV ₱50,878; Payback 1.78 yrs. Exact to the peso!
Slide 16	Conclusions	Ch. 5 (pp. 129–131)	Needs Sync	Update checkout reduction to 82.26% and reflect that SMA and ES were comparable over the 23-day window.
Slide 17	Recommendations	Ch. 4 (pp. 123–124, Table 33); Ch. 5 (pp. 132–134)	100% Match	30-to-60 day parallel run, cloud backups, 15-minute auto session timeout, and ranked enhancements (Barcode #1, Cashless #2, BIR #3).
Slide 18	Closing / Q&A Slide	Page 1	100% Match	Department and institution credits.

2. Detailed Analysis of the Three (3) Discrepancies

🚨 Discrepancy #1: Objective 3 — Forecasting Accuracy (Slide 8, 11, and 16)

The Core Issue: The slides claim that Exponential Smoothing (ES) was superior and outperformed SMA, whereas the final approved manuscript demonstrates that 3-day SMA had slightly lower error and concludes that H1 was NOT supported over the 23-day historical extraction period.

❌ CURRENT SLIDE 11 TEXT

SMA MAPE: 22.85% | **ES MAPE:** 8.40%

MAE: 7.37 ➔ 2.92 (ES won)

RMSE: 9.94 ➔ 4.38

"ES recorded lower error metrics, confirming hypothesis."

✅ FINAL MANUSCRIPT TABLE 32 (P. 122) & CONCLUSION (P. 131)

3-Day SMA Baseline: MAE = **0.52** | RMSE = **0.85** | MAPE = **92.78%**

Exponential Smoothing (α=0.7): MAE = **0.54** | RMSE = **0.92** | MAPE = **94.68%**

"Consequently, the hypothesis that Exponential Smoothing yields a lower MAE than the SMA baseline is not supported over this short-term window."

💡 ***Suggested Defense Talking Point:***

"Over our 23-calendar-day historical window (July 10 to August 1, 2026), both algorithms demonstrated comparable forecasting error rates, with the 3-day SMA yielding an MAE of 0.52 compared to 0.54 for Exponential Smoothing (α = 0.7). Because of the short timeframe and intermittent zero-demand days, the hypothesis that Exponential Smoothing yields a lower MAE was not supported. However, both models successfully provided functional, non-negative restocking suggestions for the proprietors without requiring complex historical infrastructure."

⚠️ Discrepancy #2: Objective 1 — Checkout Processing Speed (Slide 8 & 9)

The Core Issue: Slides 8 and 9 display older preliminary test data (84.15 seconds, 60.07% reduction), whereas Table 30 on Page 119 of the final manuscript records the final time-motion study showing a much faster 37.38 seconds (82.26% reduction).

❌ CURRENT SLIDE 8 & 9 TEXT

Manual Checkout: 210.76s (SD 44.30)

Digital Checkout: 84.15s (SD 12.20)

Reduction: 60.07% (126.61s faster)

t = 27.12, p < 0.001

✅ FINAL MANUSCRIPT TABLE 30 (P. 119) & CONCLUSION 1 (P. 129)

Manual Processing: 210.76s (SD 44.30)

Web POS Processing: 37.38s (SD = 12.57)

Reduction: 82.26% (**173.38s** faster)

t = 27.12, p < 0.001 (Statistically Significant)

> 💡 ***Action Needed:*** *Update Slide 8 and Slide 9 with **37.38s** and **82.26%**. This presents an even stronger narrative for your system, cutting customer wait times by nearly 3.5 minutes per transaction!*

⚠️ Discrepancy #3: Objective 5 — ISO/IEC 25010 Security Score (Slide 13)

The Core Issue: Slide 13 lists the Security rating as 3.26 (and displays an older S-CVI note of 0.78), whereas Table 21 on Page 110 of the manuscript shows a General Weighted Mean of 3.14 (with an overall instrument S-CVI of 1.00 in Table 14).

❌ CURRENT SLIDE 13 TEXT

Overall Quality Mean: 3.60 / 4.00

Functional Suitability: 3.90 / 4.00

Security Score: 3.26 / 4.00 (S-CVI = 0.78)

✅ FINAL MANUSCRIPT TABLE 21 (P. 110) & SUMMARY (P. 128)

Overall Grand Mean: 3.60 / 4.00

Functional Suitability: 3.90 / 4.00

Security Mean: 3.14 / 4.00 (Admin Override Item 3 = 2.30)

Content Validity Index (S-CVI): 1.00 (Table 14, p. 103)

💡 ***Why Item 3 Scored 2.30:*** *IT experts observed that administrative privilege overrides were not separated from the main cashier login in Phase 1. You should explain that this led directly to your technical recommendation in Chapter 5 to add a 15-minute inactivity session timeout and stricter RBAC boundaries before multi-user deployment.*

3. Cost-Benefit Analysis (CBA) Financial Verification

The financial model in Slide 15 matches Tables 10 through 13 of Chapter 3 with 100% mathematical precision:

FINANCIAL METRIC	PRESENTATION (SLIDE 15)	MANUSCRIPT (CH. 3, TABLES 10–13)	STATUS
Total Initial Investment (CapEx)	₱85,800	₱85,800 (Table 11, p. 61)	Exact Match
Hardware Cost (Existing PC/Laptop)	₱0	₱0 (Table 11, p. 60)	Exact Match

FINANCIAL METRIC	PRESENTATION (SLIDE 15)	MANUSCRIPT (CH. 3, TABLES 10–13)	STATUS
Software (SSL, Hosting, Domain, Storage × 3 Yrs)	₱30,000	₱30,000 (₱10,000/yr × 3) (Table 11, p. 61)	Exact Match
Subsidized Development Labor	₱50,800	₱50,800 (Table 10, p. 60)	Exact Match
Miscellaneous (Utilities & Documentation)	₱5,000	₱5,000 (Table 11, p. 61)	Exact Match
Annual Tangible Benefits	₱48,320 / yr	₱48,320 (₱48,000 spoilage + ₱320 logbooks) (p. 63)	Exact Match
3-Year Projected Tangible Benefits	₱144,960	₱144,960 (₱48,320 × 3 yrs) (Table 12, p. 63)	Exact Match
3-Year Net Benefit	₱59,160	₱59,160 (₱144,960 - ₱85,800) (Table 13, p. 64)	Exact Match
Return on Investment (ROI)	68.95%	68.95% (Table 13, p. 65)	Exact Match
Benefit-Cost Ratio (BCR)	1.69 : 1	1.69 : 1 (Table 13, p. 65)	Exact Match
Net Present Value (NPV @ 3% discount rate)	₱50,878	₱50,878 (Table 13, p. 65)	Exact Match
Simple Payback Period	1.78 years (~21.3 mos)	1.78 years (~21.3 months) (p. 65, 129)	Exact Match

4. Quick Action Checklist for Slide Updates

To ensure 100% alignment before defending, make these quick edits directly in your Google Slides:

- ✓ **Slide 8 (Summary Table):** Update Checkout Duration to `210.76s → 37.38s (82.26% reduction)`. Update Forecasting to `SMA MAE = 0.52 vs ES MAE = 0.54`.
- ✓ **Slide 9 (Objective 1):** Change Digital POS Mean from `84.15s` to **`37.38s`**, and change reduction from `60.07%` to **`82.26%`**.

- ✓ **Slide 11 (Objective 3):** Replace metrics with Table 32: `3-day SMA (MAE 0.52, RMSE 0.85, MAPE 92.78%)` vs. `ES α=0.7 (MAE 0.54, RMSE 0.92, MAPE 94.68%)`. State that both provided functional reorder guidance.
- ✓ **Slide 13 (Objective 5):** Update Security Score from `3.26` to **`3.14`**.
- ✓ **Slide 16 (Conclusion):** Update checkout reduction bullet to **82.26%**. Clarify that both SMA and Exponential Smoothing generated usable reorder guidance without requiring complex data architectures.

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